

Name: _____

These questions assess your understanding of the material from Chapters 1, 2 of Openstax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 50 fs. What is the time in seconds? Report your answer in scientific notation.

ANSWER: _____

2. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is 8.41×10^{10} m. What is the distance in gigameters? Report your answer in scientific notation.

ANSWER: _____

3. Classify PF_3 as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

4. Provide the number of protons and the number of neutrons there are in an atom of ^{20}Ne .

ANSWER: _____

5. Determine whether zinc and sulfur can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

6. Andrew must note the concentration of a solution of $(\text{H}_3\text{O})_2\text{O}$ in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Andrew weighs an empty graduated cylinder and finds its mass to be 3.2 g. He then puts some solid $(\text{H}_3\text{O})_2\text{O}$ into the graduated cylinder and weighs it. With the $(\text{H}_3\text{O})_2\text{O}$ added, the cylinder weighs 8.353 g. He adds water to the graduated cylinder and dissolves the $(\text{H}_3\text{O})_2\text{O}$ completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 58.15 mL. What concentration should Andrew write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

7. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$68. \frac{\text{mol}}{\text{L}} \times 95.9 \text{ L}$$

ANSWER: _____

8. As part of her duties as an analytical chemist, Olivia measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. She then records these masses in a table in her lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Olivia would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	18.4 g	23.0 g
2	5.9 g	7.4 g
3	9.6 g	12.0 g
4	9.2 g	11.5 g

ANSWER: _____

9. A botanist measures the thickness of a leaf. The thickness is 8.2 mm. What is the thickness in meters? Report your answer in standard notation, as a decimal.

ANSWER: _____

10. Grace measures the mass of an aluminum cylinder using a balance. The accepted mass of the object is 15.9 g. She records a measurement of 15.15 g. What is the percent error of her measurement? Report your answer to two significant figures.
ANSWER: _____
11. A cell biologist measures the width of a cell. The width is 58 μm . What is the width in meters? Report your answer in standard notation, as a decimal.
ANSWER: _____
12. An engineer measures the capacity of an electricity-generating plant. The capacity is 473 MW. What is the capacity in watts? Report your answer in scientific notation.
ANSWER: _____
13. An optician measures the diameter of her patient's pupil. The diameter is 0.0030 m. What is the diameter in millimeters? Report your answer in scientific notation.
ANSWER: _____
14. Jeremy needs to find the volume of a 20.55 g sample of a concentrated solution. The density of the sample is $3.0 \frac{\text{g}}{\text{mL}}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.
ANSWER: _____
15. Jacob measures the speed of a moving object and records $110.5 \frac{\text{km}}{\text{h}}$. Convert this speed to $\frac{\text{m}}{\text{s}}$. Ensure your answer has the correct number of significant figures.
ANSWER: _____
16. A European researcher reports an experimental temperature for a U.S. publication. The temperature is 25°C . Convert this temperature to degrees Fahrenheit and round to the nearest integer.
ANSWER: _____
17. What multiplicative factor does the metric prefix 'milli' (symbol: m) represent?
ANSWER: _____
18. Write the name of the inorganic acid H_2CO_3 .
ANSWER: _____
19. Allison looks up the density of a material and finds it listed as $4.7 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.
ANSWER: _____
20. Provide the name and empirical formula of the ionic compound that is formed by the Cr^{3+} cation and the SO_3^{2-} anion.
ANSWER: _____
21. Jason needs to know the mass of a sample of $\text{Ba}(\text{ClO}_2)_2$ to 2 significant figures. He puts the sample on a digital scale. This is what the scale shows: 104.66 g. If this measurement is precise enough for Jason, round it to 2 significant figures and report your answer in scientific notation. Remember to include units in your answer.
ANSWER: _____
22. Provide the chemical formula for the molecular compound ammonia.
ANSWER: _____
23. Provide the chemical symbol of the most commonly formed ion of copper and indicate whether it is a cation or an anion. Make sure you include its charge.
ANSWER: _____
24. Answer the following questions: is Ba^{2+} an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?
ANSWER: _____
25. A nurse measures the weight of his patient's medicine. The weight is 0.0031 g. What is the weight in milligrams? Report your answer in standard notation, as a decimal.
ANSWER: _____

26. A geophysicist measures the pressure in a borehole. The pressure is 8.233×10^7 Pa. What is the pressure in megapascals? Report your answer in standard notation, as a decimal.

ANSWER: _____

27. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$540 \text{ mol} \div 340 \text{ L}$$

ANSWER: _____

28. A geologist measures the height of a mountain. The height is 4.0721 km. What is the height in meters? Report your answer in scientific notation.

ANSWER: _____

29. Identify whether the following describes a random error or a systematic error, and explain your answer: A thermometer is poorly calibrated and reads 2.0°C too high at every temperature.

ANSWER: _____

30. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is 2.6509×10^{-7} m. What is the wavelength in nanometers? Report your answer in standard notation, as a decimal.

ANSWER: _____

31. As part of her duties as an analytical chemist, Olivia measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. She then records these masses in a table in her lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Olivia would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	18.4 g	23.0 g
2	5.9 g	7.4 g
3	9.6 g	12.0 g
4	9.2 g	11.5 g

ANSWER: _____

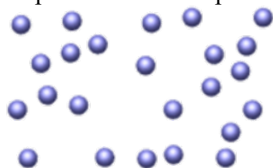
32. Classify the density of lead as a physical property or a chemical property. Explain your answer.

ANSWER: _____

33. State the SI base unit used to measure amount of substance and give its symbol.

ANSWER: _____

34. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? Explain why you chose your answer.



ANSWER: _____

35. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$0.1897 \text{ cm} + 0.5572 \text{ cm}$$

ANSWER: _____

36. Provide the name and empirical formula of an ionic compound that is formed by the aluminum cation and the sulfate anion.

ANSWER: _____

37. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is 7.05×10^{-14} m. What is the wavelength in femtometers? Report your answer in standard notation, as a decimal.

ANSWER: _____

38. A millworker measures the width of a plank. The width is 5.6 dm. What is the width in meters? Report your answer in standard notation, as a decimal.

ANSWER: _____

39. A materials scientist monitors a laboratory furnace. The furnace temperature is 81°C. Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

ANSWER: _____

40. As part of his duties as an analytical chemist, Eric analyzes a sample of an unknown pure substance X and finds that it contains 7.6 g of element E₁ and 8.9 g of element E₂, and no other elements. He then analyzes three other substances that also contain only elements E₁ and E₂ and records the results in a table in his lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E₁	mass of E₂
X	7.6 g	8.9 g
A	16.7 g	6.5 g
B	27.7 g	5.4 g
C	50.7 g	9.9 g

ANSWER: _____